

Andrew Ortegaray

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EDUCATION

University of California, Santa Barbara (UCSB) <i>Ph.D. in Mathematics, Advisor: Zhenghan Wang</i>	Santa Barbara, CA <i>Sep. 2024 – Present</i>
University of California, Santa Barbara (UCSB) <i>M.A. in Mathematics</i>	Santa Barbara, CA <i>June 2025</i>
California Institute of Technology (Caltech) <i>Bachelor of Science in Mathematics and Physics, Advisor: Matilde Marcolli</i>	Pasadena, CA <i>Sep. 2016 – June 2020</i>

RESEARCH INTERESTS

Quantum Topology, Topological Quantum Computation, Quantum Error Correction, Topological Data Analysis

PAPERS AND PUBLICATIONS

Heat Kernel Analysis of Syntactic Structures (with R. Berwick and M. Marcolli), *Mathematics in Computer Science*, Vol. 15 (2021), 643–660.

Phylogenetics of Indo-European Language Families via an Algebro-Geometric Analysis of Their Syntactic Structures (with K. Shu, R. Berwick, and M. Marcolli), *Mathematics in Computer Science*, Vol. 15 (2021), 803–857.

An Implementation of Persistent Homology, *Wolfram Summer Program* (2020).

RESEARCH EXPERIENCE

Wolfram Research <i>Symbolic Algorithms Researcher</i>	Seattle, WA <i>Aug. 2020 – Aug. 2021</i>
Fast Number Theoretic Transforms (FNTTs): Implemented FFT-style algorithms for computing FNTTs over number fields in the Wolfram Language. Developed fast matrix-based methods for small transforms and divide-and-conquer strategies for larger inputs. Implementations are published and available in current Wolfram Language distributions.	
Fast Multipole Methods (FMMs): Designed FMMs for efficient evaluation of many-particle potentials in 1D, 2D, and 3D. Achieved $O(n)$ evaluation complexity compared to default $O(n^2)$ methods, with practical performance crossover at $n > 8000$.	
Wolfram Summer Program <i>Developer</i>	Champaign, IL <i>June 2020 – Aug. 2020</i>
Developed Wolfram Language methods to construct Vietoris-Rips complexes from point cloud data and compute persistent homology, including access to homology generators. Implementation is available in current Wolfram Language distributions. See publications above.	
Summer Undergraduate Research Fellowship <i>Research Fellow</i>	Pasadena, CA <i>June 2016 – June 2017</i>
Analyzed binary vectors representing the presence or absence of syntactic structures in languages using statistical, algebro-geometric, and topological methods. Used nearest-neighbor methods to estimate statistical interdependence among syntactic structures. Constructed and reproduced phylogenetic trees of different language families. See publications above.	

TEACHING EXPERIENCE

UCSB <i>Teaching Assistant</i>	Santa Barbara, CA <i>Sep. 2024 – Present</i>
Led discussion sections, office hours, and exam reviews. Developed course materials and assessments. Courses: MA 3A/3B/34B (Single Var. Calculus), MA 4B (Diff. Eqs.), MA 6B (Multivar. Calculus)	
Caltech <i>Teaching Assistant</i>	Pasadena, CA <i>Mar. 2018 – Dec. 2019</i>
Managed grading, prepared solution sets, and assisted with instruction. Courses: Ma 0 (Intro. to Proofs), Ma 5 (Group Theory), Ph 1 (Classical Mechanics and Electromagnetism), and Ph 2 (Quantum and Statistical Mechanics)	
Art of Problem Solving Academy <i>Math Instructor</i>	Bellevue, WA <i>Sep. 2022 – Sep. 2024</i>
Led advanced extracurricular mathematics courses ranging from pre-calculus to AMC 12-level competition math. Classes typically included up to 15 students of diverse ages and backgrounds.	
UPchieve <i>Math and Science Coach</i>	Seattle, WA <i>Jan. 2023 – Sep. 2024</i>
Assisted students regularly with high school-level mathematics and science coursework. Emphasized conceptual understanding, critical thinking, and open-ended problem solving.	

MENTORING AND SERVICE

UCSB REU <i>Undergraduate Mentor</i>	Santa Barbara, CA <i>June 2025 – Oct. 2025</i>
Guided a multi-student REU in quantum error correction.	
Women in STEM <i>Undergraduate Mentor</i>	Santa Barbara, CA <i>Jan. 2025 – Present</i>
Provided professional mentorship for undergraduate women in STEM fields.	
Directed Reading Program <i>Project Mentor</i>	Santa Barbara, CA <i>Jan. 2025 – Present</i>
Guided undergraduate reading projects, primarily in topology.	
Youth Math League <i>Lead Coach</i>	Arcadia, CA <i>Jan. 2019 – June 2020</i>
Organized a youth math program to prepare advanced 5th/6th grade girls for Olympiad-style competitions. Organized and led weekly practices and lessons.	

INDUSTRY EXPERIENCE

SpaceX <i>Starlink Analyst</i>	Seattle, WA <i>Sep. 2022 – Sep. 2024</i>
Developed monitoring and analytics tools for assessing Starlink network health across orbital assets, terrestrial interconnections, broadcast sites, and global points of presence. Integrated heterogeneous data sources from multiple subsystem teams into scalable operational metrics for live network performance. Designed health and performance metrics to support rapid network growth and improve diagnosis of capacity, isolation, and service-impacting events.	

TECHNICAL SKILLS

Python, Wolfram Language, Mathematica, C++, R, Lean, Linux, Git, $\text{\LaTeX}$, symbolic computation, numerical methods, topological data analysis